Pandelis Zembashis

CTO | Systems Engineer | Agent Runtime, Performance & Developer Tools

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EXECUTIVE SUMMARY

CTO and hands-on systems engineer building the runtime layer underneath AI coding agents. At Cosine, built and led work on a single Go harness shared by CLI, desktop, VS Code/WebAssembly, API-server sessions, and Kubernetes worker pods: streaming tool calls, event scheduling, checkpointing, deterministic replay, repository-scale search, sandboxed execution, and production debugging. Strong fit for Claude Code Systems through agent-loop architecture, runtime performance work, Linux/container interfaces, Rust/Go systems programming, JavaScript/TypeScript product depth, and direct OSS work around Claude and Codex SDKs.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – May 2026

AI coding lab building agent runtimes, developer tools, model feedback loops, and secure execution infrastructure.

- Led product and engineering for Cosine's AI coding platform, setting technical direction across agent runtime architecture, tool execution, context handling, performance, reliability, and developer experience.
- Built the core Go harness for streaming tool calls, event scheduling, cancellation, checkpointing, deterministic replay, memory writes, MCP/LSP integrations, subagents, file locks, and structured rollout traces.
- Optimized repository and filesystem hot paths around native Go concurrency, bundled ripgrep-class search, streaming file reads, low-overhead tool results, and predictable behavior on large codebases.
- Shipped one runtime across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, keeping execution semantics stable across local, embedded, browser-adjacent, and cloud-backed surfaces.
- Built remote execution for model-controlled coding workloads using Kubernetes workers, gVisor/runsc isolation, ephemeral sessions, WebSocket control planes, secure shell/tool access, and network boundaries.
- Built debugging and evaluation machinery around real agent runs: replayable traces, test results, LSP diagnostics, checkpoints, reviews, conversation patterns, and parallel sandboxed executions.
- Worked across OpenAI, Anthropic, Gemini, and internal model capabilities, turning provider-specific behavior into stable prompts, tool contracts, context strategies, and runtime-level product behavior.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled LLM-backed outbound automation systems that processed millions of email chains and billions of tokens, with event-driven architecture, worker orchestration, and operational monitoring.
- Established engineering cadence, incident response habits, onboarding, and operating rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a production fleet management platform in Next.js, React Query, Tailwind CSS, and TypeScript.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving iteration across web, operations, and marketplace workflows.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

SELECTED OSS, WRITING & DEVELOPER TOOLING

Claude/Codex SDK Work: claude-agent-sdk-rust; claude-agent-sdk-go; codex-web-sdk; OpenAI-compatible middleware.
Cosine Agent Systems: Why we built our own harness; Cosine Swarm; Secure AI coding agents for enterprise.
Developer Tooling: Cosine CLI 1.16: built-in search, single-binary distribution, OS pipes, Go/Rust integration, and ripgrep-level internal search architecture.
Public OSS Signal: github.com/PandelisZ: 100+ merged public GitHub PRs across developer tooling, infrastructure, SDKs, and product systems.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Agent Runtime Systems: Tool execution, event loops, checkpoints, replay, subagents, MCP/LSP, context handling, long-running task state
Systems & Performance: Go, Rust, Linux, containers, WebAssembly, filesystem/search hot paths, tracing, profiling, memory and latency reduction
Developer Tooling: TypeScript, Node.js, React, Next.js, GraphQL, CLI tooling, SDK design, terminal-first workflows, IDE integrations
Secure Execution: Kubernetes, gVisor/runsc, ephemeral sandboxes, WebSockets, shell/tool access, tenant isolation, VPC/on-prem deployment
Technical Leadership: Staff-level architecture, product/runtime tradeoffs, mentoring, incident response, research-product collaboration